

Practical Applications of Measurements and Transformational Geometry

Measurement is one of the most practical areas of mathematics. Every day, people measure lengths, calculate areas, estimate volumes, and plan layouts without even thinking of it as mathematics. A carpenter measures wood before cutting. A tailor measures fabric before sewing. A homeowner calculates how much paint is needed for a wall. A farmer estimates how much rice a storage container can hold. A family packing a balik-bayan box needs to know if everything will fit. These are all applications of measurement. Alongside measurement, transformational geometry helps us understand how shapes can be moved, flipped, turned, and resized – skills used in design, pattern-making, tiling, and layout planning.

This reading material covers the practical applications of measurements and transformational geometry. It explains how to estimate and measure lengths and distances, calculate the area and perimeter of residential spaces, find the volume and capacity of three-dimensional objects, and apply volume and capacity in cost estimation.

Part 1: Estimating and Measuring Lengths and Distances

Why Estimation Matters

Before measuring precisely, it is useful to **estimate** – to make a reasonable guess based on known references. Estimation helps you decide how much material to buy, whether furniture will fit in a room, or how much fabric a project requires. Good estimators then verify with actual measurement.

Common Units of Length

Unit	Abbreviation	Equivalent
Millimeter	mm	0.001 m
Centimeter	cm	0.01 m
Meter	m	Base unit
Kilometer	km	1,000 m
Inch	in	2.54 cm
Foot	ft	30.48 cm
Yard	yd	0.9144 m

Converting Units

To convert between metric units, multiply or divide by powers of 10.

From	To	Operation
cm	m	Divide by 100
m	cm	Multiply by 100
m	km	Divide by 1,000

From	To	Operation
km	m	Multiply by 1,000
mm	cm	Divide by 10
cm	mm	Multiply by 10

Example 1: Converting Units

A fabric is 250 cm long. How many meters is this?

$$250 \text{ cm} \div 100 = 2.5 \text{ m}$$

Answer: The fabric is **2.5 meters** long.

Measuring Floor Dimensions

When measuring a room, you need the **length** and **width**. Use a tape measure and record in meters or centimeters.

Example 2: Floor Dimensions

A bedroom measures 4.2 m long and 3.5 m wide. What is the total floor area?

$$\text{Area} = \text{Length} \times \text{Width}$$

$$\text{Area} = 4.2 \text{ m} \times 3.5 \text{ m}$$

$$\text{Area} = 14.7 \text{ m}^2$$

Answer: The floor area is **14.7 square meters**.

Measuring Fabric for Clothing

Fabric is often sold by the meter or yard. A tailor must estimate how much fabric is needed based on the garment.

Example 3: Fabric for a Blouse

A blouse requires 1.5 m of fabric for the body and 0.5 m for the sleeves. If the tailor wants a 10% allowance for cutting errors, how much fabric should be bought?

Total needed = $1.5 + 0.5 = 2.0$ m

Allowance = 10% of 2.0 = 0.2 m

Total to buy = $2.0 + 0.2 = 2.2$ m

Answer: The tailor should buy **2.2 meters** of fabric.

Part 2: Area and Perimeter of Residential Spaces

Perimeter and Area Formulas

Shape	Perimeter	Area
Rectangle	$P = 2(L + W)$	$A = L \times W$
Square	$P = 4s$	$A = s^2$
Triangle	$P = a + b + c$	$A = \frac{1}{2} \times \text{base} \times \text{height}$
Circle	$C = 2\pi r$	$A = \pi r^2$
Trapezoid	$P = a + b + c + d$	$A = \frac{1}{2}(a + b)h$

Room Renovations

Example 1: Painting a Room

A room is 5 m long, 4 m wide, and 3 m high. You want to paint the four walls (not the ceiling or floor). One liter of paint covers 12 m^2 . How many liters are needed for one coat?

Step 1: Find the wall area.

Two walls: $5 \text{ m} \times 3 \text{ m} = 15 \text{ m}^2$ each $\rightarrow 30 \text{ m}^2$

Two walls: $4 \text{ m} \times 3 \text{ m} = 12 \text{ m}^2$ each $\rightarrow 24 \text{ m}^2$

Total wall area = $30 + 24 = 54 \text{ m}^2$

Step 2: Divide by coverage per liter.

Liters needed = $54 \div 12 = 4.5 \text{ L}$

Answer: You need **4.5 liters** of paint for one coat.

Example 2: Tiling a Floor

A bathroom floor is 2.5 m by 2 m. Tiles are $0.3 \text{ m} \times 0.3 \text{ m}$. How many tiles are needed?

Step 1: Find the floor area.

Area = $2.5 \times 2 = 5 \text{ m}^2$

Step 2: Find the area of one tile.

Tile area = $0.3 \times 0.3 = 0.09 \text{ m}^2$

Step 3: Divide.

Number of tiles = $5 \div 0.09 \approx 55.56$

Answer: You need at least **56 tiles** (round up).

Garden Layouts

Example 3: Rectangular Garden

A garden is 8 m long and 5 m wide. You want to fence it and plant grass.

Perimeter (fencing):

$$P = 2(8 + 5) = 2(13) = 26 \text{ m}$$

Area (grass):

$$A = 8 \times 5 = 40 \text{ m}^2$$

Answer: You need **26 meters** of fencing and **40 square meters** of grass.

Example 4: Circular Garden

A circular garden has a radius of 3 m. Find its circumference and area.

$$\text{Circumference} = 2\pi r = 2 \times 3.1416 \times 3 \approx 18.85 \text{ m}$$

$$\text{Area} = \pi r^2 = 3.1416 \times 9 \approx 28.27 \text{ m}^2$$

Answer: Circumference $\approx$ **18.85 m**; Area $\approx$ **28.27 m²**

Composite Shapes

Many residential spaces are not simple rectangles. Break them into smaller shapes.

Example 5: L-Shaped Room

An L-shaped room can be split into two rectangles:

- Rectangle A: $6 \text{ m} \times 4 \text{ m} = 24 \text{ m}^2$
- Rectangle B: $3 \text{ m} \times 2 \text{ m} = 6 \text{ m}^2$
- Total area = $24 + 6 = 30 \text{ m}^2$

Part 3: Volume and Capacity of Three-Dimensional Objects

Volume vs. Capacity

Term	Definition	Units
Volume	The amount of space an object occupies	cm ³ , m ³
Capacity	The amount a container can hold	mL, L

Key conversions:

1 cm³ = 1 mL
1 m³ = 1,000 L
1 L = 1,000 mL

Volume Formulas

Solid	Formula
Rectangular prism (box)	$V = L \times W \times H$
Cube	$V = s^3$
Cylinder	$V = \pi r^2 h$
Pyramid	$V = \frac{1}{3} \times \text{base area} \times h$

Solid

Formula

Cone

$$V = \frac{1}{3}\pi r^2 h$$

Sphere

$$V = \frac{4}{3}\pi r^3$$

Water Tanks

Example 1: Cylindrical Water Tank

A cylindrical water tank has a radius of 1.5 m and a height of 3 m. How many liters can it hold?

Step 1: Find the volume in m^3 .

$$V = \pi r^2 h$$

$$V = 3.1416 \times (1.5)^2 \times 3$$

$$V = 3.1416 \times 2.25 \times 3$$

$$V \approx 21.21 \text{ m}^3$$

Step 2: Convert to liters.

$$21.21 \text{ m}^3 \times 1,000 = 21,210 \text{ L}$$

Answer: The tank can hold approximately **21,210 liters**.

Rice Storage Containers

Example 2: Rectangular Rice Container

A rice storage container is 1.2 m long, 0.8 m wide, and 0.6 m high. How much rice can it hold in liters?

$$V = 1.2 \times 0.8 \times 0.6 = 0.576 \text{ m}^3$$

$$\text{Capacity} = 0.576 \times 1,000 = 576 \text{ L}$$

Answer: The container can hold **576 liters** of rice.

Balik-Bayan Boxes

Example 3: Balik-Bayan Box

A balik-bayan box measures 60 cm × 40 cm × 35 cm. Find its volume in cm³ and in liters.

$$V = 60 \times 40 \times 35 = 84,000 \text{ cm}^3$$

$$\text{Capacity} = 84,000 \text{ cm}^3 = 84,000 \text{ mL} = 84 \text{ L}$$

Answer: The box has a volume of **84,000 cm³** and a capacity of **84 liters**.

Composite Containers

Some containers are combinations of shapes. Add the volumes together.

Example 4: Water Tank with Cylindrical Top

A tank has a rectangular base 2 m × 1.5 m × 1 m and a cylindrical top with radius 0.75 m and height 0.5 m.

$$\text{Rectangular part: } 2 \times 1.5 \times 1 = 3 \text{ m}^3$$

$$\text{Cylindrical part: } \pi(0.75)^2(0.5) \approx 0.884 \text{ m}^3$$

$$\text{Total} = 3 + 0.884 = 3.884 \text{ m}^3 \approx \mathbf{3,884 \text{ liters}}$$

Part 4: Applying Volume and Capacity in Cost Estimation

The Basic Cost Formula

$$\text{Total Cost} = \text{Quantity Needed} \times \text{Unit Price}$$

Add a **waste allowance** (often 5–10%) for cutting, spillage, or errors.

Baking: Flour for Bread Production

Example 1: Flour for Bread

A bakery produces 200 loaves of bread per day. Each loaf requires 500 g of flour. How much flour is needed per day, and what is the cost if flour is ₱60 per kilogram?

$$\text{Flour per day} = 200 \times 500 \text{ g} = 100,000 \text{ g} = 100 \text{ kg}$$

$$\text{Cost} = 100 \text{ kg} \times \text{₱}60 = \text{₱}6,000$$

Answer: The bakery needs **100 kg** of flour per day, costing **₱6,000**.

Example 2: Scaling a Recipe

A recipe for 12 muffins needs 300 g of flour. How much flour is needed for 60 muffins?

$$\text{Flour per muffin} = 300 \div 12 = 25 \text{ g}$$

$$\text{For 60 muffins} = 60 \times 25 = 1,500 \text{ g} = 1.5 \text{ kg}$$

Answer: You need **1.5 kg** of flour.

Painting: Paint Needed for Walls

Example 3: Paint for a Room

A room has a total wall area of 72 m². One liter of paint covers 12 m². If paint costs ₱350 per liter, how much will one coat cost?

$$\text{Liters needed} = 72 \div 12 = 6 \text{ L}$$

$$\text{Cost} = 6 \times \text{₱}350 = \text{₱}2,100$$

Answer: One coat costs **₱2,100**.

Example 4: Two Coats with Waste Allowance

For two coats with a 10% allowance:

$$\text{Total paint} = 6 \text{ L} \times 2 \text{ coats} = 12 \text{ L}$$

$$\text{With 10\% allowance} = 12 \times 1.10 = 13.2 \text{ L}$$

$$\text{Cost} = 13.2 \times \text{₱}350 = \text{₱}4,620$$

Answer: Two coats with allowance cost **₱4,620**.

Concrete for a Slab

Example 5: Concrete Volume

A concrete slab is 4 m long, 3 m wide, and 0.1 m thick. How much concrete is needed?

$V = 4 \times 3 \times 0.1 = 1.2 \text{ m}^3$

Answer: You need **1.2 cubic meters** of concrete.

Part 5: Transformational Geometry in Practical Design

Transformational geometry studies how shapes change position or size through **transformations**. These are used in design, layout, and pattern-making.

Transformation	What It Does	Practical Example
Translation (slide)	Moves a shape without rotating or flipping	Tiling a floor with the same tile pattern
Reflection (flip)	Flips a shape over a line	Symmetrical garden design; mirror patterns in fabric
Rotation (turn)	Turns a shape around a point	Rotating a furniture layout; fan blade design
Dilation (resize)	Changes the size of a shape proportionally	Scaling a floor plan to fit paper; enlarging a dress pattern
Glide reflection	Slide + flip	Wallpaper and border patterns

Applying Transformations to Layouts

Example 1: Tiling a Floor

A square tile is repeated across a floor using **translation**. The same tile is slid horizontally and vertically to cover the area without gaps. This is called a **tessellation**.

Example 2: Symmetrical Garden

A garden path is designed with **reflection symmetry**. The left half mirrors the right half, creating balance and visual appeal.

Example 3: Scaling a Floor Plan

A floor plan is 10 m × 8 m. To fit it on a sheet of paper, you use **dilation** with a scale of 1:100.

Paper dimensions = $10 \text{ m} \div 100 = 0.10 \text{ m} = 10 \text{ cm}$

Paper dimensions = $8 \text{ m} \div 100 = 0.08 \text{ m} = 8 \text{ cm}$

Answer: The floor plan fits on a **10 cm × 8 cm** rectangle.

Key Terms

Term	Definition
Estimation	Making a reasonable guess based on known references
Perimeter	The total distance around a shape
Area	The amount of surface a shape covers
Volume	The amount of space an object occupies
Capacity	The amount a container can hold
Composite shape	A shape made by combining two or more simple shapes
Waste allowance	Extra material added to cover cutting, spillage, or errors
Tessellation	A pattern of shapes that fit together without gaps or overlaps
Transformation	A change in position or size of a shape
Dilation	A transformation that changes size proportionally

Review Questions

1. Convert 3,500 cm to meters.
2. A rectangular room is 6 m long and 4.5 m wide. Find its area and perimeter.
3. A circular garden has a radius of 4 m. Find its circumference and area.
4. A cylindrical water tank has a radius of 2 m and a height of 5 m. Find its volume in m^3 and its capacity in liters.
5. A balik-bayan box measures $70 \text{ cm} \times 50 \text{ cm} \times 40 \text{ cm}$. Find its volume in cm^3 and its capacity in liters.
6. A room has a wall area of 90 m^2 . One liter of paint covers 15 m^2 . Paint costs ₱400 per liter. How much will two coats cost, including a 10% waste allowance?
7. A bakery produces 150 loaves per day. Each loaf needs 450 g of flour. Flour costs ₱55 per kilogram. Find the daily flour cost.
8. An L-shaped room is made of two rectangles: $7 \text{ m} \times 5 \text{ m}$ and $3 \text{ m} \times 2 \text{ m}$. Find the total area.
9. A floor plan of a $12 \text{ m} \times 9 \text{ m}$ house is drawn at a scale of 1:100. What are the dimensions on paper?
10. Application question: A rectangular rice storage container is 1.5 m long, 1 m wide, and 0.8 m high.
 - (a) Find its volume in m^3 .
 - (b) Find its capacity in liters.
 - (c) If rice costs ₱50 per kilogram and 1 liter of rice weighs approximately 0.8 kg, find the total value of rice that can fill the container.

Answer Key

1. $3,500 \div 100 = \mathbf{35\ m}$

2. Area = $6 \times 4.5 = \mathbf{27\ m^2}$; Perimeter = $2(6 + 4.5) = \mathbf{21\ m}$

3. $C = 2\pi(4) \approx \mathbf{25.13\ m}$; $A = \pi(4)^2 \approx \mathbf{50.27\ m^2}$

4. $V = \pi(2)^2(5) \approx \mathbf{62.83\ m^3}$; Capacity = $\mathbf{62,830\ L}$

5. $V = 70 \times 50 \times 40 = \mathbf{140,000\ cm^3}$; Capacity = $\mathbf{140\ L}$

6. Liters = $90 \div 15 = 6\ \text{L per coat}$; two coats = $12\ \text{L}$; with 10% = $13.2\ \text{L}$; Cost = $13.2 \times \text{P}400 = \mathbf{\text{P}5,280}$

7. Flour = $150 \times 450 = 67,500\ \text{g} = 67.5\ \text{kg}$; Cost = $67.5 \times \text{P}55 = \mathbf{\text{P}3,712.50}$

8. $(7 \times 5) + (3 \times 2) = 35 + 6 = \mathbf{41\ m^2}$

9. $12\ \text{m} \div 100 = \mathbf{12\ cm}$; $9\ \text{m} \div 100 = \mathbf{9\ cm}$; paper size $\mathbf{12\ cm \times 9\ cm}$

10.

(a) $V = 1.5 \times 1 \times 0.8 = \mathbf{1.2\ m^3}$

(b) Capacity = $1.2 \times 1,000 = \mathbf{1,200\ L}$

(c) Weight = $1,200 \times 0.8 = 960\ \text{kg}$; Value = $960 \times \text{P}50 = \mathbf{\text{P}48,000}$